

Paediatric Medicine

Child Health, Growth, Development & Common Paediatric Conditions

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1. Normal Child Development

Developmental surveillance is integral to paediatric primary care. Milestones are population-based medians — there is wide normal variation. Significant deviation or regression requires prompt assessment.

Age	Gross Motor	Fine Motor	Language	Social/Cognitive
2 months	Lifts head prone	Hands unfisted	Coos	Social smile
6 months	Sits with support	Transfers objects	Babbles, turns to voice	Recognises parents
12 months	Pulls to stand, cruises	Pincer grasp	1–2 words (mama/dada)	Waves bye-bye, plays peek-a-boo
18 months	Walks independently	Stacks 2–3 blocks	10–20 words	Parallel play, points to show
2 years	Runs, kicks ball	Draws vertical line	2-word phrases (50+ words)	Symbolic play
3 years	Climbs stairs alternate feet	Draws circle, tripod grip	3-word sentences (250 words)	Group play, knows name/age/gender
5 years	Skips, hops	Draws person (6 parts)	Complex sentences; tells stories	Follows rules, understands cause-effect

■ **RED FLAGS** requiring immediate referral: Regression of milestones at any age; no social smile by 3 months; no babbling by 12 months; no words by 16 months; no 2-word phrases by 24 months; loss of language at any age.

2. Fever in Children

2.1 Fever Assessment by Age

Fever (core temperature $\geq 38.0^{\circ}\text{C}$) is one of the most common reasons for paediatric presentation. Risk stratification depends critically on age, clinical appearance, and associated symptoms.

Age Group	Key Concern	Management Approach
< 28 days (neonate)	Presumed serious bacterial infection (sepsis, meningitis, UTI)	Full sepsis workup + IV antibiotics; hospital admission
1–3 months	Serious bacterial infection risk remains high but lower	Full workup if unwell; selective admission; low threshold for blood cultures
3–36 months	Viral illness most common; UTI common in girls	Clinical assessment (NICE traffic light); urinalysis; targeted investigations
> 3 years	Viral URI, tonsillitis, otitis media most common	Supportive care; antibiotics only if bacterial source identified

■ NICE Traffic Light System: Green (likely viral, manage at home with safety netting), Amber (uncertain — senior review, further investigation), Red (seriously unwell — immediate treatment/admission). Do not rely on temperature height alone.

3. Common Paediatric Conditions

3.1 Bronchiolitis

Bronchiolitis is the most common lower respiratory tract infection in infants, predominantly caused by Respiratory Syncytial Virus (RSV). It typically affects children under 2 years, with peak incidence in winter months.

- Prodrome: 1–3 days of coryza and fever
- Progressive lower respiratory symptoms: wheeze, cough, increased work of breathing
- Signs: subcostal/intercostal recession, nasal flaring, tachypnoea, fine crackles and wheeze
- Apnoea risk in young infants (< 2 months) and premature infants
- Treatment: Supportive — oxygen for $\text{SpO}_2 < 90\%$, IV fluids/NG feeds if unable to feed, positioning

■ Admit if: $\text{SpO}_2 < 92\%$ on air, moderate-severe respiratory distress, apnoea, poor feeding (< 50% normal intake), age < 3 months or premature, significant comorbidity, or parental concern about safety at home.

3.2 Otitis Media (Acute)

Acute otitis media (AOM) is a bacterial or viral middle ear infection common in young children due to anatomical differences in Eustachian tube orientation. *S. pneumoniae*, *H. influenzae*, and *M. catarrhalis* are

the main bacterial pathogens.

- Otolgia (ear pain), tugging at ear in infants
- Fever (may be high), irritability and sleep disturbance
- Otoscopy: red/dull/bulging tympanic membrane, loss of light reflex, limited mobility
- Management: Analgesia (paracetamol/ibuprofen); antibiotics if < 2 yrs, bilateral AOM, severe symptoms, or no improvement at 48–72 h
- Amoxicillin 40–90 mg/kg/day in 3 divided doses for 5–10 days (first-line)

3.3 Febrile Convulsions

Febrile seizures occur in 2–5% of children aged 6 months to 5 years in association with fever. They are the most common convulsive disorder in childhood and are generally benign.

Feature	Simple Febrile Seizure	Complex Febrile Seizure
Duration	< 15 minutes	> 15 minutes
Type	Generalised (tonic-clonic)	Focal onset or generalised
Frequency (24h)	Single episode	Recurr within 24 hours
Post-ictal	Brief (< 1 hour)	Prolonged or focal deficit
Risk of Epilepsy	Slightly above baseline (~2–3%)	Higher risk (~6–8%)
Management	Observation; treat fever; parent education	Investigate; consider LP; neurology referral

■ **Manage active febrile seizure: Place in recovery position; protect from injury; time the seizure; if > 5 minutes — administer rescue benzodiazepine (rectal diazepam or buccal midazolam); call emergency services for status epilepticus (> 30 min).**

4. Child Immunisation Schedule (Summary)

The Expanded Programme on Immunisation (EPI) schedule protects against vaccine-preventable diseases from birth. Timing is critical for optimal immunogenicity.

Age	Vaccines
Birth	BCG, Hepatitis B (1st dose), Oral Polio Vaccine (OPV-0)
6 weeks	DTwP/DTaP-1, IPV-1, Hib-1, Hepatitis B-2, PCV-1, Rota-1
10 weeks	DTwP/DTaP-2, IPV-2, Hib-2, PCV-2, Rota-2

Age	Vaccines
14 weeks	DTwP/DTaP-3, IPV-3, Hib-3, Hepatitis B-3, PCV-3
6 months	Influenza-1 (then annual)
9 months	Measles (MR/MMR-1) in some programmes
12–15 months	MMR-1, Varicella-1, PCV booster, Hepatitis A-1
15–18 months	DTwP/DTaP booster, IPV booster, Hib booster
4–6 years	DTaP booster, IPV booster, MMR-2, Varicella-2

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